Week10 (11/20-11/26)

MoonWalker Weekly Meeting Slides

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MoonWalker Weekly Meeting

11/19 - 11/26 Summary

Elijah, Rowel, Martin

Assignments from Dr. Lu

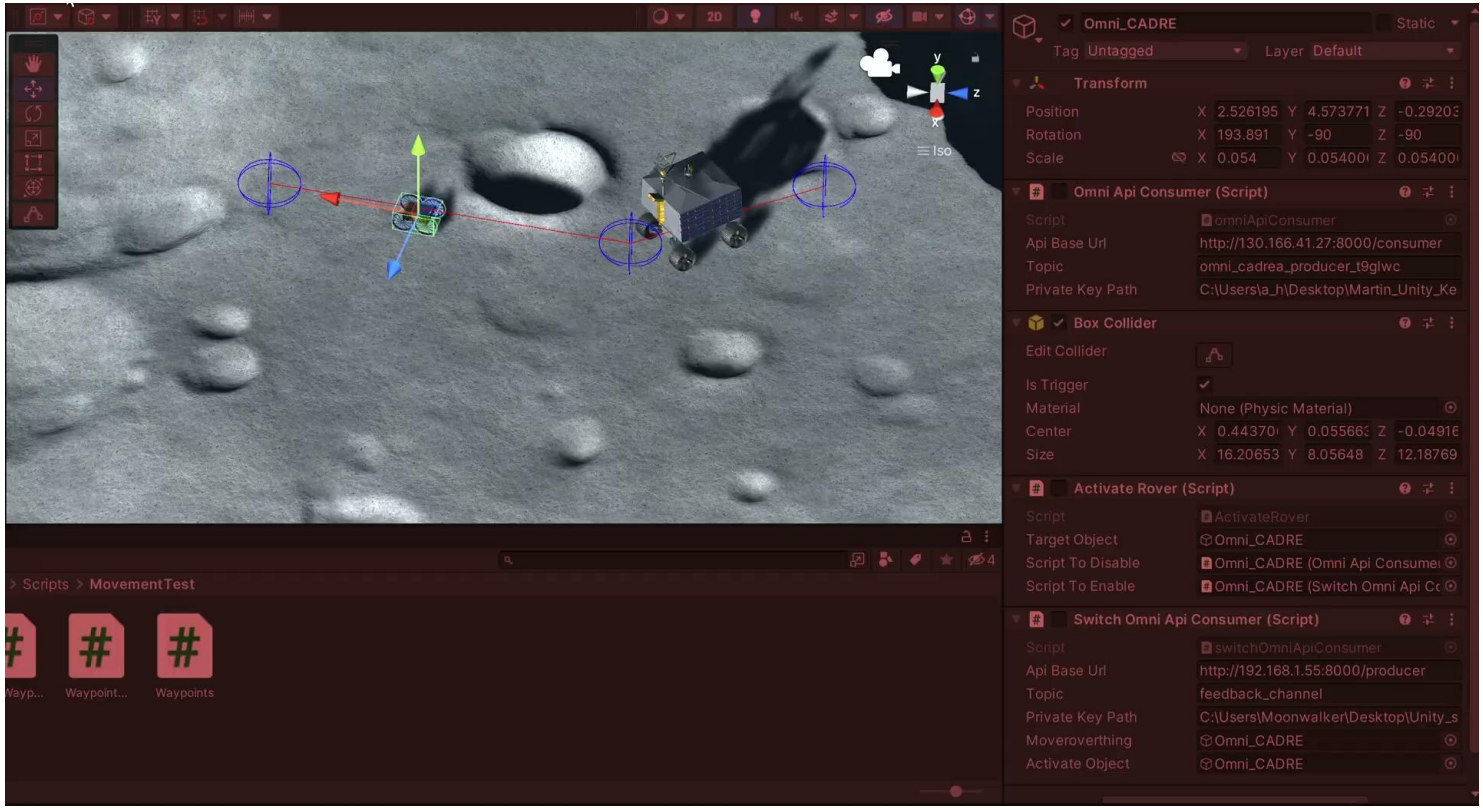
Over the next two weeks:

- Figure out how to toggle between hosting
- Perform local collision tests
  - See how crashes look in each simulation
- Figure out how to send physics and which metrics
  - Starting with mass is okay

Progress Update

- Martin
  - Implemented Unity-hosting toggle upon collision
  - Identified possible physics metrics to send/receive
    - Mass
    - Drag (affects positional/movement speed)
    - Angular Drag (affects rotational speed)
- Elijah
  - Created script to switch from producer to consumer for Omniverse. Script includes enabling kinematics mode for Cadre, making it into a ghost on omniverse. Able to switch back and forth
    - Still need to implement collision detection into it
  -
- Rowel
  - Began Updating HSML Schema (hsml.jsonID) and Schema Properties.xlsx
    - Added inControl(boolean) and mass(number, kg) under Agents
    - Work with team to add further physics parameters

Consumer to Producer Switching (Unity)



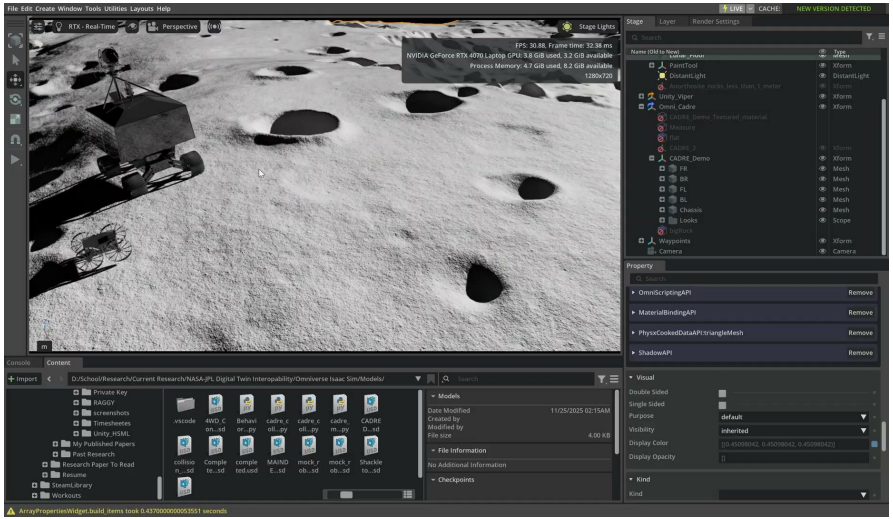
Example JSON from Unity

- rb stands for the rigidbody added to Cadre at moment of Collision
- Why these metrics?
  - Mass, drag, and angular drag are values that remain constant.
  - Omniverse version of the “drag” metrics should be called linear damping & angular damping

- The idea is that we choose a platform to handle the physics at the beginning and initialize once with these values for the rover.

```
{ "additionalProperty", new JArray {
    new JObject { { "@type", "schema:PropertyValue" }, { "name", "scale" }, { "value", 100 } },
    new JObject { { "@type", "schema:PropertyValue" }, { "name", "mass" }, { "value", rb != null ? rb.mass : 0 } },
    new JObject { { "@type", "schema:PropertyValue" }, { "name", "drag" }, { "value", rb != null ? rb.drag : 0 } },
    new JObject { { "@type", "schema:PropertyValue" }, { "name", "angularDrag" }, { "value", rb != null ? rb.angularDrag : 0 } }
}},
```

Producer to Consumer Switching (Omniverse)



Kinematic Switch Script

Producer Consumer Switch Script

Issues with ConsumerProducer Script

- When stopping simulation after switching from producer to consumer, the chassis of Cadre is not getting re-initialized to the right rotation and translation
- Cause: Orient was not being restored to (0.0, 0.0, 0.0)
- Solution explored:
  - Saving initial state of Cadre on on_init()

```
# Keep initial values
if self.translate_op:
    pos_val = self.translate_op.Get()
    if pos_val is None:
        pos_val = Gf.Vec3d(0.0, 0.0, 0.0)
        self.translate_op.Set(pos_val)
    self.initial_local_pos = pos_val
else:
    self.initial_local_pos = Gf.Vec3d(0.0, 0.0, 0.0)

if self.orient_op:
    rot_val = self.orient_op.Get()
    if rot_val is None:
        rot_val = Gf.Quatd(1.0, 0.0, 0.0, 0.0)
        self.orient_op.Set(rot_val)
    self.initial_local_rot = rot_val
else:
    self.initial_local_rot = Gf.Quatf(1.0, 0.0, 0.0, 0.0)
```

File Edit Create Window Tools Utilities Layouts Help
LIVE
CACHE:
NEW VERSION DETECTED
RTX - Real-Time
Perspective
Stage Lights
FPS: 46.06, Frame time: 21.71 ms
NVIDIA GeForce RTX 4070 Laptop GPU: 3.7 GiB used, 3.3 GiB available
Process Memory: 8.1 GiB used, 7.8 GiB available
1280x720
Stage
Layer
Render Settings
Search
Name (Old to New)
Type
PaintTool
Xform
DistantLight
DistantLight
Anorthosite_rocks_less_than_1_meter
Xform
Unity_Viper
Xform
Omni_Cadre
Xform
CADRE_Demo_Textured_material
Measure
flat
CADRE_2
Xform
CADRE_Demo
Xform
FR
Mesh
BR
Mesh
FL
Mesh
BL
Mesh
Chassis
Mesh
Looks
Scope
bigRock
Waypoints
Xform
Camera
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Property
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Add
Lunar_Floor
Prim Path
/World/Shackleton_Ridge_300x300_v4./Lunar_Floor
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Transform
Translate
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Y 0.0
Z 1128.86353
Rotate
X 0.0
Y 0.0
Z 0.0
Scale
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Y 39.37008
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Materials on selected models
Geometry
Mesh
Normals
Orientation
rightHanded
Points
Console
Content
Import
D:/School/Research/Current Research/NASA-JPL Digital Twin Interopability/Omniverse Isaac Sim/Models/
Search
Models
.vscode
Scripts
Setup
Setup Documenta
Papers
Private Key
RAGGY
screenshots
Timesheetes
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4WD_Con...sd
Behavior...py
cadre_coll...py
cadre_coll...py
cadre_m...py
CADRED...sd
collision_...sd
Complete...sd
completed.usd
MAINDE...sd
mock_rob...sd
mock_rob...sd
Shackleto...sd
Models
Date Modified
11/26/2025 02:19AM
Created by
Modified by
File size
0.00 KB
File Information
No Additional Information
Checkpoints
ArrayPropertiesWidget.build_items took 0.28100000001722947 seconds

Action Plan

- Rowel
  - Look at Martin and Elijah’s scripts and update the API accordingly
  - Begin testing API/JSON
- Elijah
  - Include automatic collision detection in the Consumer Producer Switch script
  - Include other physics parameters in the JSON file being produced in the Consumer Producer Switch script
  - Work with Rowel & Martin on implementing hsml schema changes
    - Based on JSON files being written by Omniverse
- Martin
  - Work with Rowel & Elijah on implementing hsml schema changes.
    - Based on JSON files being written by Unity
  - Work with Rowel regarding feedback channel for on-collision switch.